

Bachelor's Thesis

Quantifying Labelling Uncertainty in Image Data

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Abstract

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1 Introduction

2 Data

2.1 Introduction to the Datasets

We will analyse labelling uncertainty in the context of image classification using the *CIFAR-10H* dataset. The dataset is based on the *CIFAR-10* (Canadian Institute for Advanced Research, 10 classes) dataset, first introduced in Krizhevsky (2009), which is a benchmark dataset for image classification and is itself a subset of the larger *80 million tiny images* dataset, see Torralba et al. (2008). The *CIFAR-10* dataset consists of 60'000 images of size 32×32 pixels. Each image belongs to one of ten classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck. Some example images of each class are shown in Figure 1. The images in the *CIFAR-10* dataset come with multiple advantages for our purposes, see Peterson et al. (2019). First, the dataset is well-known and widely used in the machine learning community, see *Papers with Code - CIFAR-10 Dataset* (2024) for a list of papers that use the dataset. Second, the low resolution of the images makes the dataset challenging for image classification and introduces ambiguity in the labelling process. Third, the dataset contains images that are close to the category boundaries, which further introduces ambiguity in the labelling process. This is in contrast with other datasets that are more carefully created such that each image is selected to be a unambiguously of a certain category.

The *CIFAR-10* dataset is divided into a training set of 50'000 images and a test set of 10'000 images. The test set is the basis for the *CIFAR-10H* dataset, which will be used in this thesis. *CIFAR-10H* is a balanced dataset, meaning that each class contains the same number of images, i.e. 1000 images. *CIFAR-10H* is a human-labelled dataset, where each image has been labelled by multiple human annotators. The dataset contains 514'200 observations from a total of 2571 annotators. Each annotator labelled 200 images, 20 from each category, which were randomly sampled from the test set of the *CIFAR-10* dataset. On average, each image was labelled 51 times, with the minimum number of labels being 50 and the maximum number of labels being 63. The relevant features of the *CIFAR-10H* dataset are described in Table 1. The study design is described in more detail in Peterson et al. (2019), section 4.2. Note, that the images were collected based on searching the web, using search engines. Thus, the true label is not necessarily the object that is displayed in the image, but the term that was used to search for the image, see Torralba et al. (2008). For our purposes, we will assume that the true label is the object that is displayed in the image.

The *CIFAR-10* and *CIFAR-10H* datasets are publicly available and can be downloaded from the CIFAR-10 website *CIFAR-10 and CIFAR-100 Datasets* (n.d.) and the author's GitHub repository *Jcpeterson/Cifar-10h: Original Dataset Release for CIFAR-10H* (n.d.).

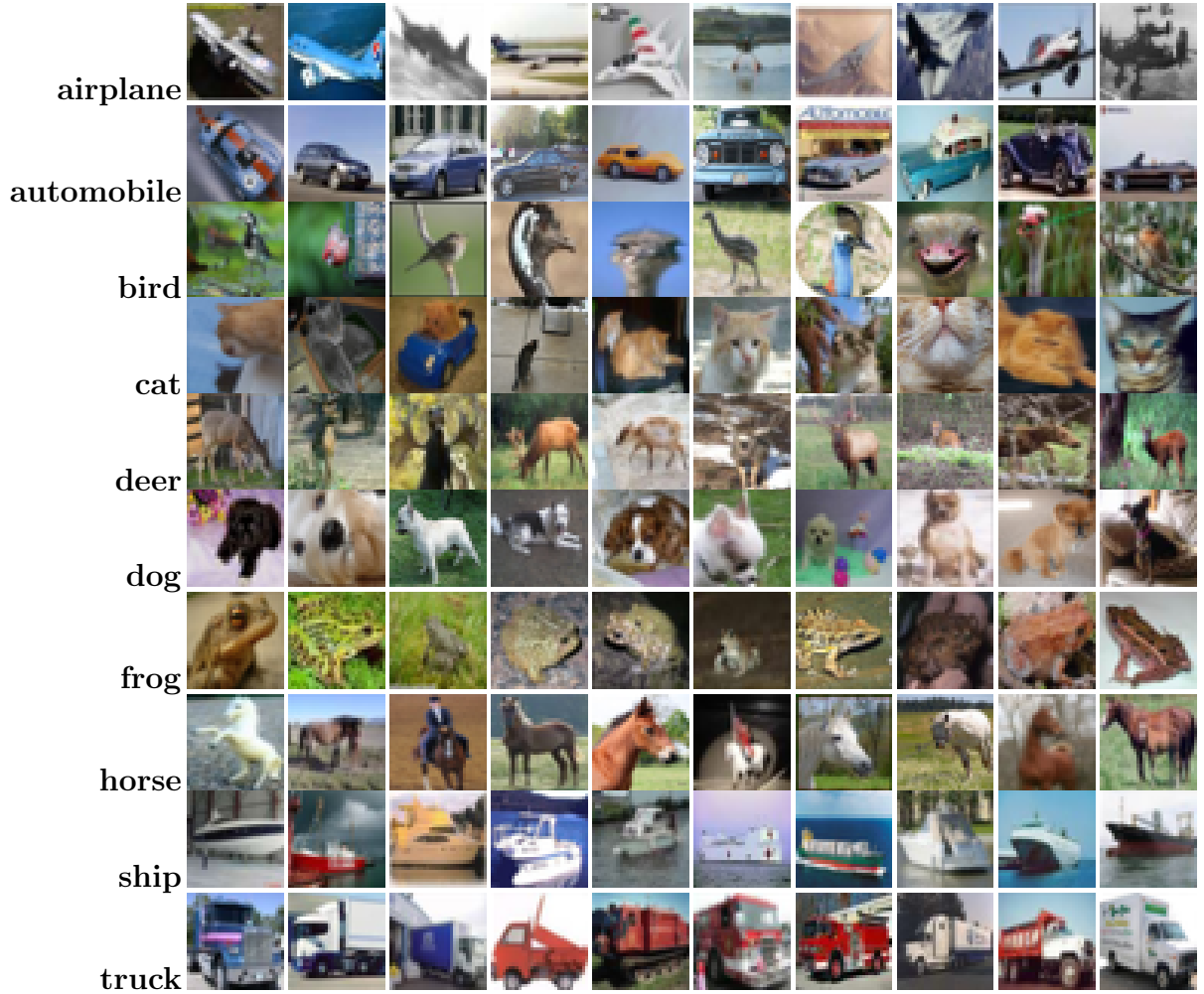


Figure 1: The classes of the CIFAR-10 dataset with ten example images per class.

Feature	Description
annotator_id	Unique integer ID ranging from 0 to 2570
trial_index	Indexes the trial for each annotator, from 0 to 219
true_category	True string class name (e.g., "cat") of the image. The set of categories is {airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck}
chosen_category	String class name (e.g., "cat") chosen by the annotator for the image from the set of possible true categories
correct_guess	Whether the guess was correct (i.e., when true_category == chosen_category), marked by 1, or 0 otherwise
image_filename	Filename of the PNG (e.g., "cabin_cruiser_s_000814.png")
subcategory	Subordinate class label, which is recoverable from the image filenames (e.g., "cabin_cruiser")
reaction_time	Time taken (in milliseconds) by the annotator to provide a response for the current trial
time_elapsed	Total time (in milliseconds) that passed at the end of each trial. The first is equal to the reaction_time for the first trial. The rest are essentially cumulative reaction_time values

Table 1: Features of the CIFAR-10H dataset.

2.2 Descriptive Statistics

2.2.1 Label Ambiguity

As mentioned in the previous section, the *CIFAR-10H* dataset is balanced with respect to the true categories. This means, that we have 51'420 observations per true category. But the number of observations per chosen category is not balanced, see Figure 2, e.g. the category *deer* is underrepresented, while the category *dog* is overrepresented. The question that arises is, whether chosen categories are usually correct or not and where the votes of overrepresented categories come from.

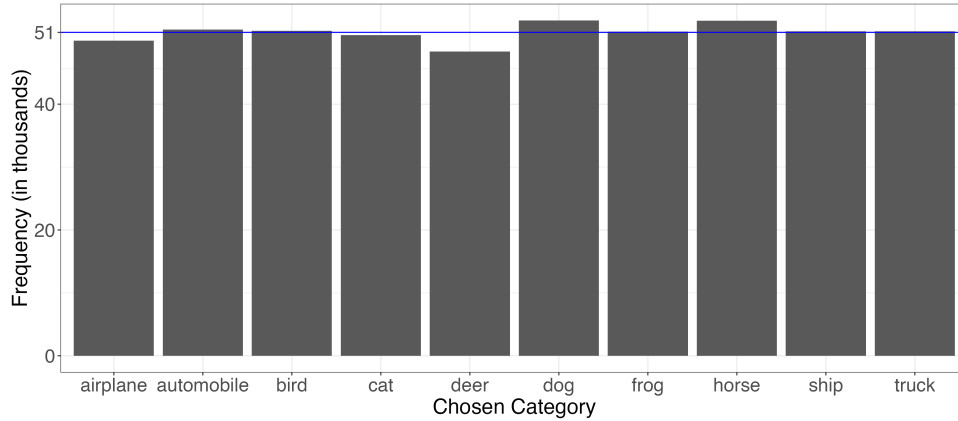


Figure 2: Number of observations per chosen category. The blue horizontal line indicates the 51'420 observations per true category, so that we can see, which categories are over- and underrepresented.

We can also question the ambiguity of the images. For that, we can check, how many unique labels each image has received. If an image is unambiguous, then all annotators should agree on the label of the image. If an image is ambiguous, then there are multiple possible labels for the image. In Figure 3, we can see the distribution of the number of labels per image for each true category. We can see, that in all categories, 50% of the images have received at most 2 labels. We can also see, that in almost all categories, 25% of the images have received 3 labels or more. This indicates that a lot of images are unambiguous, but also that a considerable amount of images are ambiguous. In Figure 4, we can see an examples of ambiguous and unambiguous images as well as their respective vote distribution.

Another possibility to check the ambiguity of the images is to check the entropy of the vote distribution. The entropy of an image is defined as

$$H = - \sum_{i=1}^{10} p_i \log_2(p_i), \text{ with } p_i = \frac{n_i}{N}, \quad (1)$$

and n_i is the number of votes for category i and N is the total number of votes that the image received. A high entropy indicates that the image is ambiguous, while a low entropy, i.e. close to 0, indicates that the image is unambiguous. In Figure 5, that is

replicated from Battleday et al. (2020), we can see the distribution of the entropy and can again confirm that most images are rather unambiguous, but that a considerable amount of images are ambiguous. Note, that the entropy can be at most 3.33, because that’s the value under the assumption that the votes for a certain image are uniformly distributed.

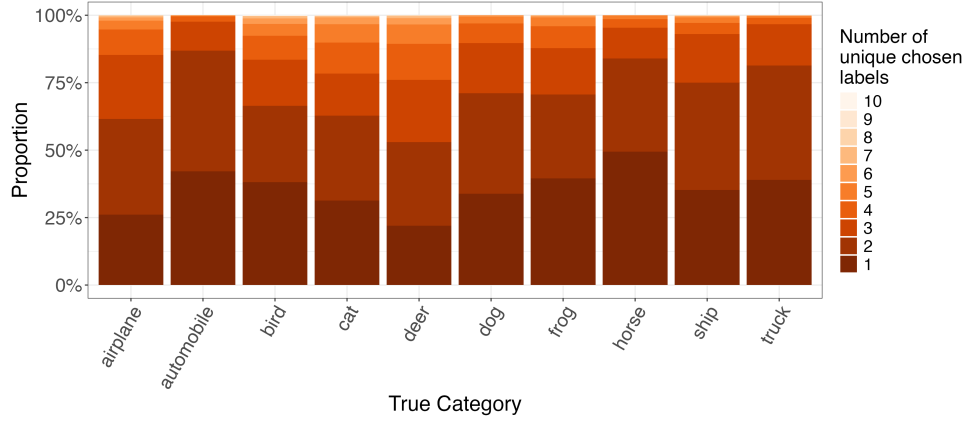


Figure 3: Distribution of the number of unique chosen labels per image for each true category. Unambiguous images have received only one label, while ambiguous images have received multiple labels. The more labels an image has received, the more ambiguous the image is.

This ambiguity in the labelling process is also reflected in the confusion matrix in Figure 6. The confusion matrix shows the proportion of votes for each chosen category given the true category. We can see that the diagonal of the confusion matrix is not 1, because the annotators do mislabel the image. We can also see that the confusion matrix is not symmetric, which indicates that the mislabelling is not symmetric, i.e. the annotators do not confuse the categories in both directions equally. For example, a deer is more often confused with a horse than a horse is confused with a deer. We can also see, that the confusion matrix is almost sparse, i.e. there are a lot of values that are almost 0, which indicates that the annotators do not confuse the categories randomly, but that there are some categories that are more often confused with each other. One goal of this thesis is to estimate the confusion matrix from Figure 6 by only using the chosen categories.

Lastly, we can check the number of subcategories per true category. The subcategory is a more detailed label of the image. The subcategories are not balanced, see Table 2. On average, there are 16 images per subcategory, but there is a lot of variance in the number of images per subcategory. The distribution of the number of images per subcategory is shown in Appendix A, Figure 11. One question of interest is, whether there are subcategories which more ambiguous images, i.e. with more labelling uncertainty, than others.

True Cat.	airp.	auto.	bird	cat	deer	dog	frog	horse	ship	truck
#Subcat.	40	35	45	38	30	42	37	39	40	31

Table 2: Number of subcategories for every true category.

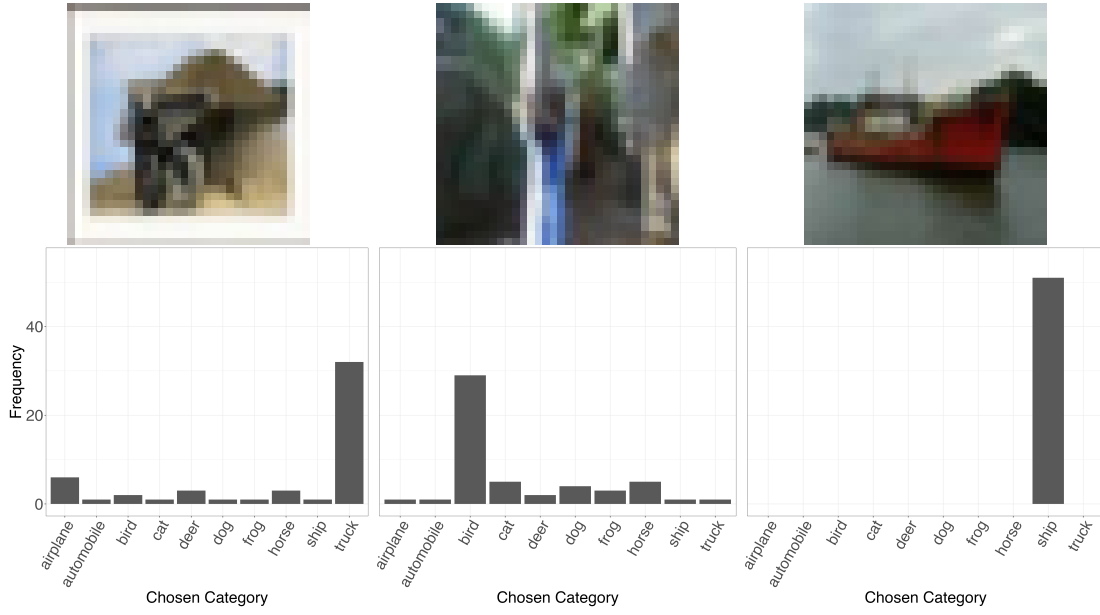


Figure 4: Example of an two ambiguous images and an unambiguous image and their respective vote distribution. The true label of the images are *truck*, *bird* and *ship*, which all received the majority of votes.

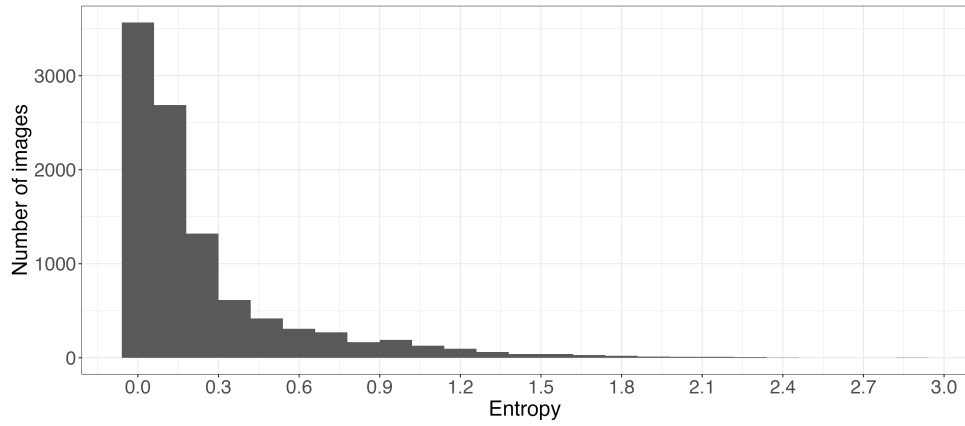


Figure 5: Distribution of the entropy of the vote distribution for each true category. Most of the images have a low entropy, which indicates that they are unambiguous. But a considerable amount of images have a high entropy, which indicates that they are ambiguous.

True Category	airplane	94.80%	0.80%	1.10%	0.20%	0.10%	0.10%	0.30%	0.10%	2.00%	0.40%
	automobile	0.30%	96.80%	0.10%	0.20%	0.10%	0.10%	0.10%	0.10%	0.10%	2.30%
	bird	0.40%	0.10%	94.40%	1.20%	1.00%	0.90%	1.30%	0.40%	0.20%	0.10%
	cat	0.20%	0.10%	1.40%	91.20%	0.80%	4.30%	1.30%	0.40%	0.20%	0.10%
	deer	0.20%	0.10%	1.10%	1.00%	90.20%	2.10%	0.70%	4.30%	0.20%	0.10%
	dog	0.10%	0.00%	0.60%	3.50%	0.40%	94.60%	0.30%	0.40%	0.10%	0.10%
	frog	0.10%	0.00%	1.20%	1.40%	0.60%	0.70%	95.80%	0.10%	0.10%	0.10%
	horse	0.10%	0.10%	0.30%	0.20%	0.80%	0.70%	0.10%	97.60%	0.10%	0.10%
	ship	1.20%	0.40%	0.20%	0.20%	0.10%	0.10%	0.20%	0.10%	96.90%	0.60%
	truck	0.20%	2.40%	0.10%	0.10%	0.10%	0.10%	0.10%	0.10%	0.40%	96.40%
			airplane	automobile	bird	cat	deer	dog	frog	horse	ship
Chosen Category											

Figure 6: Confusion matrix of the CIFAR-10H dataset. The confusion matrix shows the proportion of votes for each chosen category given the true category.

2.2.2 Reaction Time and Elapsed Time

Another interesting feature of the *CIFAR-10H* dataset is the reaction time of the annotators. The reaction time is the time that the annotator needed to provide a response for the current trial. In Figure 7, we can see the distribution of the reaction time (in seconds). The median reaction time is around 1.48 seconds.

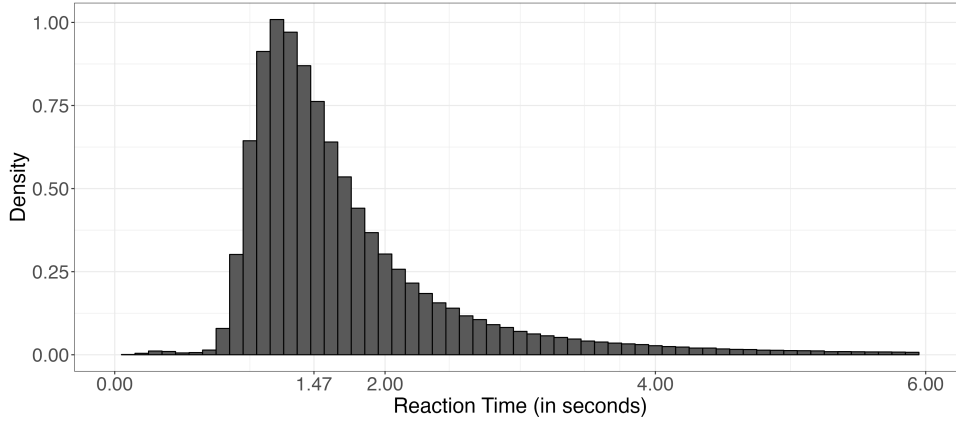


Figure 7: Distribution of the reaction time (in seconds) for each true category. The median reaction time is around 1.48 seconds. We cut off the observations that have a reaction time of more than 6 seconds, which was approx. 1.9% of the observations.

One interesting aspect of the reaction time is the relationship between the reaction time and the entropy of the vote distribution. Computing the correlation between the reaction time and the entropy of the vote distribution on the whole dataset, we get a correlation of 0.0372. But this correlation is misleading, because we have a lot of outliers in the reaction time, which are not necessarily indicative of labelling uncertainty, e.g. the longest

observed reaction time is around 2.13 hours with an image entropy of 0. We therefore need to remove the outliers from the dataset, but it is not clear, how to define an outlier in this context. In Figure 8, we can see the pearson-correlation between the reaction time and the entropy of the vote distribution based on the cutoff-value of the reaction time, beginning at 6 seconds. We can see, that the correlation is increasing first, up to a value of 0.32 at 11 seconds, and then it decreases. Using a threshold of 26 seconds for the reaction time for later analysis seems reasonable, because this is above the 99.9% quantile of the reaction time distribution. This results in a correlation of 0.30 and it is reasonable to assume that that’s enough time to label even the most ambiguous images.

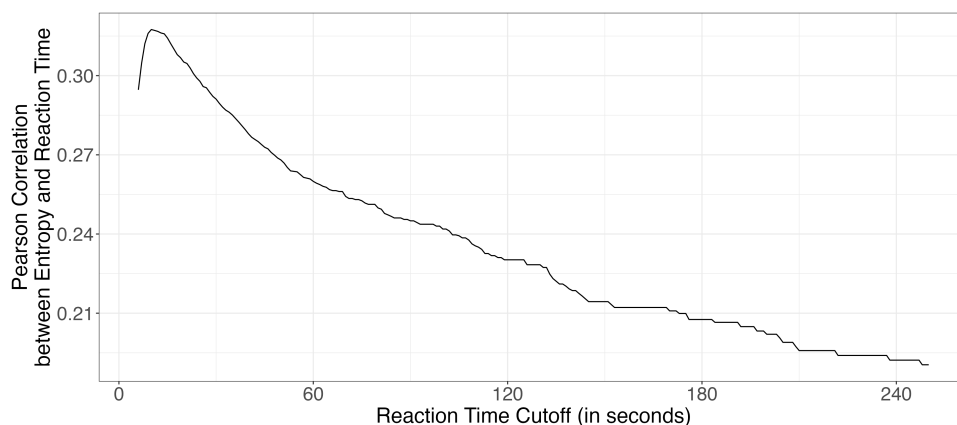


Figure 8: Pearson-correlation between the reaction time and the entropy of the vote distribution based on the cutoff-value of the reaction time.

Another interesting aspect of the reaction time is the relationship between the reaction time and the correctness of the chosen category. In Figure 9, we can see the relationship between the reaction time and the correctness of the chosen category. We can see that for a higher reaction time, the probability of a correct guess becomes lower. This will be further investigated in the next sections. For now, this is just an indicator, that a higher reaction time might be an indicator for labelling uncertainty. The question that arises is, whether the decreased probability of a correct guess is due to the fact, that some images are more ambiguous than others, thus more prone to labelling uncertainty, or whether there are annotators that are more prone to labelling uncertainty than others.

One issue that may arise in regards to the quality of the data are annotators that are not paying attention and are just randomly guessing the labels. This is supposed to be controlled for in the study design, by performing so called *attention checks*, where the annotators receive an image with a clear unambiguous label and are asked to label the image with the correct label. If the annotator failed to do so, the annotator was excluded from the study. Nonetheless, by random chance, there could still be some annotators that guessed the labels correctly and stayed in the study. In Figure 10, we can see the total elapsed time, i.e. the time that the annotator needed to label all 200 images, for each annotator (in minutes) vs. the proportion of correct guesses. We can see that the proportion of correct guesses is almost always above 80% and that the elapsed time is almost always between 5 and 20 minutes. We can also see that the annotators with a lower

proportion of correct guesses tend to need have a lower total elapsed time. Especially the annotators with a proportion of correct guesses below 25% were very fast in labelling the images, with an average total elapsed time of around 4.6 minutes, which is less than 1.4 seconds per image for all 200 images.

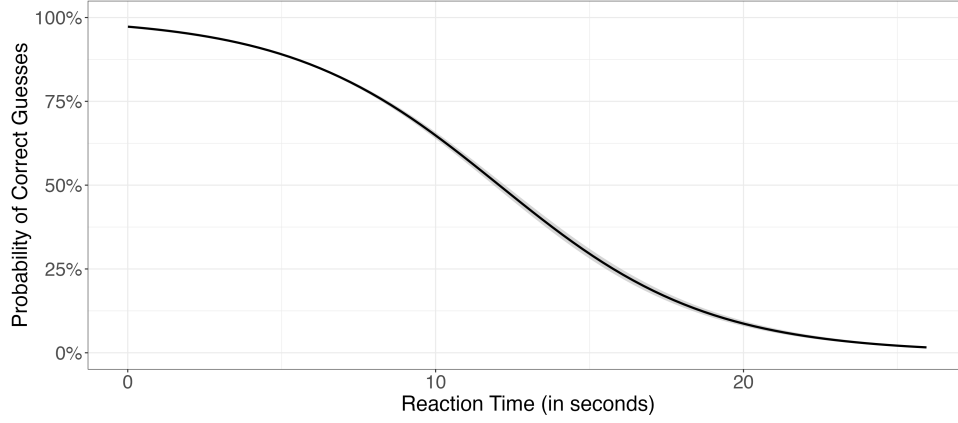


Figure 9: Average Reaction time (in seconds) for correct and incorrect guesses, based on a logistic model with reaction time as independent variable. For a higher reaction time, the probability of a correct guess is lower.

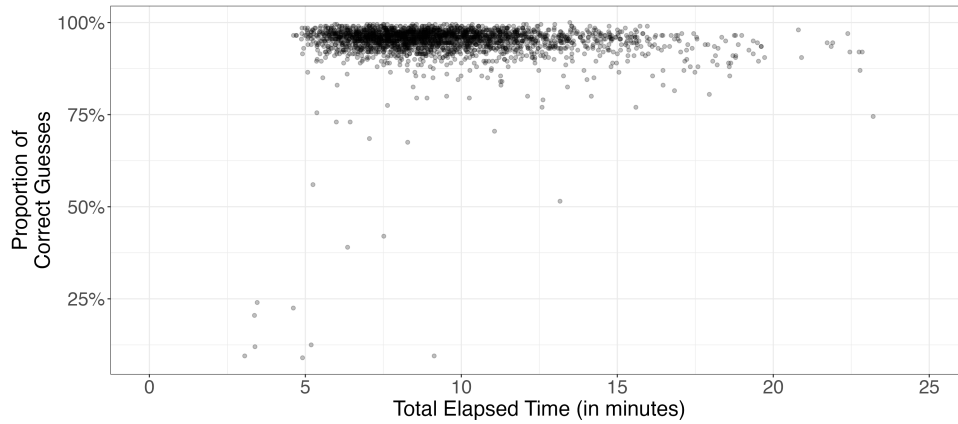


Figure 10: Total elapsed time (in minutes) for each annotator vs. the proportion of correct guesses. On average the annotators needed 9.5 minutes to label 200 images with an average proportion of correct guesses of 95%. We exclude the observations that have a total elapsed time of more than 25 minutes, which are 8 out of 2571 annotators.

3 Modelling Labelling Uncertainty

We assume that we have a setting, in which we have images that are ought to be labelled. The annotator has to choose the label from a rather small set of predefined labels. We assume that the annotator is not perfect and that there is some uncertainty in the labelling process. This uncertainty can arise from various sources such as ambiguity in the data, subjective interpretation by the annotator, and noise in the labelling process. We further assume that there is an unknown underlying ground truth to every image. Those scenarios might arise in real-world applications, like

- medical imaging: classify images of X-rays or MRIs into different medical conditions
- wildlife monitoring: classify images of animals into different species
- quality control in manufacturing: classify images of products into different quality categories
- agriculture: classify images of crops into different growth or health stages
- environmental monitoring: classify images of landscapes into different local climate zones

We base our methodology on the work of Hechinger et al. (2024), who used a Bayesian Latent Class Mixture Model to quantify the labelling uncertainty. The estimation of the parameters are done by using a Stochastic Expectation-Maximization algorithm.

3.1 Bayesian Latent Class Mixture Model

Let i with $i = 1, \dots, n$ be the index of the images and j with $j = 1, \dots, J$ be the index of the annotators. We denote the label that the annotator j assigns to the image i by $V_j^{(i)}$, with $V_j^{(i)} \in \{1, \dots, K\}$, where K is the number of classes. This leads to the definition of the accumulated votes of all annotators per image i as $\mathbf{Y}^{(i)} := (Y_1^{(i)}, \dots, Y_K^{(i)})$, with $Y_k^{(i)} := \sum_{j=1}^{J^{(i)}} I(V_j^{(i)} = k)$, $I(\cdot)$ being the indicator function and $J^{(i)}$ being the number of annotators that labelled the image i . We will refer to $\mathbf{Y}^{(i)}$ as the vote distribution of image i .

We assume, that each image i comes from a latent class, i.e. the image i belongs to a true category $Z^{(i)}$, with $Z^{(i)} \in \{1, \dots, K\}$ which is unknown (latent) and define $\mathbf{Z}^{(i)}$ as the one-hot encoded vector of $Z^{(i)}$, i.e. $\mathbf{Z}^{(i)} = (Z_1^{(i)}, \dots, Z_K^{(i)}) = (I(Z^{(i)} = 1), \dots, I(Z^{(i)} = K))$. Using this notation, we formulate our model as follows:

$$\mathbf{Z}^{(i)} \sim \text{Multi}(\boldsymbol{\pi}, 1), i.i.d. \quad \text{with} \quad \boldsymbol{\pi} = (\pi_1, \dots, \pi_K) \quad \text{and} \quad \sum_{k=1}^K \pi_k = 1, \quad (2)$$

$$\mathbf{Y}^{(i)} | \mathbf{Z}^{(i)} \sim \text{Multi}(\boldsymbol{\theta}_{\mathbf{Z}^{(i)}}, J^{(i)}) \quad \text{with} \quad \boldsymbol{\theta}_z = (\theta_{z1}, \dots, \theta_{zK}) \quad \text{and} \quad \sum_{k=1}^K \theta_{zk} = 1. \quad (3)$$

We call π_k the prior probability of class k , i.e. the probability that an image belongs to class k without knowing anything about the image. The vector $\boldsymbol{\theta}_z$ contains the probabilities $\theta_{zk}, k = 1, \dots, K$ of the annotators assigning the label k to an image of class z . We can

write the $\boldsymbol{\theta}_k$ vectors as rows of a matrix Θ , where $\Theta := (\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_K) = \begin{pmatrix} - & \boldsymbol{\theta}_1 & - \\ & \vdots & \\ - & \boldsymbol{\theta}_K & - \end{pmatrix}$,

which we call the confusion matrix. In Figure 6 we have seen an example of this confusion matrix. Given a latent class $Z^{(i)}$ and the confusion matrix Θ , we have a specific distribution for the votes $\mathbf{Y}^{(i)}$, i.e. the observed given labels come from a mixture of different distributions, based on the latent class of the image. We obtain the likelihood contribution of the image i through

$$\mathbb{P}(\mathbf{Y}^{(i)}; \boldsymbol{\pi}, \Theta) = \sum_{z=1}^K \mathbb{P}(\mathbf{Y}^{(i)} | Z^{(i)} = z; \Theta) \mathbb{P}(Z^{(i)} = z; \boldsymbol{\pi}) = \sum_{z=1}^K \pi_z \mathbb{P}(\mathbf{Y}^{(i)}; \boldsymbol{\theta}_z). \quad (4)$$

In order to estimate the parameters $\boldsymbol{\pi}$ and Θ , we use the Stochastic Expectation Maximization algorithm, which is described in the following section.

3.2 Stochastic Expectation Maximization Algorithm

Using the Expectation Maximization algorithm (Dempster et al. (1977)), we can estimate the parameters of the model. The EM algorithm is an iterative method that alternates between the E-step and the M-step. In the E-step, we compute the expected value of $\mathbf{Z}^{(i)}$ given the observed data $\mathbf{Y}^{(i)}$ and the current estimate of the parameters $\hat{\boldsymbol{\pi}}$ and $\hat{\Theta}$. This results in a complete dataset, that can then be used to estimate the best parameters in the M-step. The new estimates of the parameters are then used in the next iteration of the E-step.

The stochastic expectation maximization (SEM) algorithm (Celeux et al. (1996)) is a stochastic version of the EM algorithm, which is especially useful, when one is also interested in the uncertainty of the estimates. The SEM algorithm is based on the idea of the EM algorithm, but instead of computing the expected value of $\mathbf{Z}^{(i)}$, it samples $\mathbf{Z}^{(i)}$ from the posterior distribution.

The SEM algorithm works as follows:

1. Initialize the parameters $\hat{\boldsymbol{\pi}}_0$ and $\hat{\Theta}_0$. If no prior knowledge is available, one can initialize $\hat{\boldsymbol{\pi}}_0 = (\frac{1}{K}, \dots, \frac{1}{K})$ and draw $\hat{\Theta}_0$ randomly from a Dirichlet distribution with parameters K and $\alpha = (2K, \dots, 2K)$.
2. Repeat for $t = 0, \dots, T$:
 - (a) E-Step: Compute

$$\hat{\tau}_{z,t}^{(i)} = \mathbb{P}(Z_t^{(i)} = z | \mathbf{Y}^{(i)}; \hat{\boldsymbol{\pi}}_t, \hat{\Theta}_t) = \frac{\hat{\pi}_{z,t} \mathbb{P}(\mathbf{Y}^{(i)}; \hat{\boldsymbol{\theta}}_{z,t})}{\sum_{z'=1}^K \hat{\pi}_{z',t} \mathbb{P}(\mathbf{Y}^{(i)}; \hat{\boldsymbol{\theta}}_{z',t})}. \quad (5)$$

Use the $\hat{\tau}_t^{(i)}$ to randomly sample $\mathbf{Z}_t^{(i)} \sim \text{Multi}(\hat{\boldsymbol{\tau}}_t^{(i)}, 1)$ for all $i = 1, \dots, n$.

- (b) Update the parameter estimates $\hat{\pi}$ and $\hat{\Theta}$ using the sampled $\mathbf{Z}^{(i)}$. We use the MLE for $\hat{\pi}$ and $\hat{\Theta}$ to update the estimates. The MLE for $\hat{\pi}$ is given by

$$\hat{\pi}_{t+1} = \frac{1}{n} \sum_{i=1}^n \mathbf{Z}_k^{(i)} \quad (6)$$

$$\hat{\theta}_{zk,t+1} = \frac{\sum_{i=1}^n I(Z_t^{(i)} = z) Y_k^{(i)}}{\sum_{i=1}^n I(Z_t^{(i)} = z) J^{(i)}}. \quad (7)$$

3. Compute the final estimates $\hat{\pi}$, $\hat{\Theta}$ and $\hat{\tau}$, by averaging over the samples. Note, that we discard the first t_0 samples to avoid the burn-in phase, where the algorithm has not yet converged to the true distribution.

$$\hat{\pi} = \frac{1}{T - t_0} \sum_{t=t_0+1}^T \hat{\pi}_t \quad (8)$$

$$\hat{\Theta} = \frac{1}{T - t_0} \sum_{t=t_0+1}^T \hat{\Theta}_t \quad (9)$$

$$\hat{\tau}^{(i)} = \frac{1}{T - t_0} \sum_{t=t_0+1}^T \hat{\tau}_t^{(i)}. \quad (10)$$

The equations above can be interpreted as follows:

- Equation 5 computes the posterior probability that the image i belongs to class z , given the observed votes $\mathbf{Y}^{(i)}$ and current estimates of the parameters $\hat{\pi}_t$ and $\hat{\Theta}_t$.
- Equation 6 computes the relative frequency of the latent classes.
- Equation 7 computes how many images of latent class z have been labelled with label k , compared to the total number of votes that the latent class z received. This can also be expressed in matrix notation, see Appendix A equation 15.

Mixture models usually have an identifiability problem, i.e. the latent classes are not uniquely identifiable. This means, that we don't know which latent class $z = 1, \dots, K$ corresponds to which label $k \in \{\text{airplane, automobile, } \dots, \text{truck}\}$. To solve this problem, we can use the fact that

$$\mathbb{P}(Z^{(i)} = z | V^{(i)} = k) \propto \mathbb{P}(V^{(i)} = k | Z^{(i)} = z) \mathbb{P}(Z^{(i)} = z)$$

and use the estimated confusion matrix $\hat{\Theta}$ and the estimated latent class probabilities $\hat{\pi}$ to compute the posterior probabilities $\mathbb{P}(Z^{(i)} = z | V^{(i)} = k)$ for all $z = 1, \dots, K$ and $k = 1, \dots, K$. Based on that, we can then assign the labels to the latent classes and permute the rows of the confusion matrix $\hat{\Theta}$ for better interpretability.

Using the chain of estimates $(\hat{\pi}_t)_{t=t_0, \dots, T}$, $(\hat{\Theta}_t)_{t=t_0, \dots, T}$ and $(\hat{\tau}_t)_{t=t_0, \dots, T}$ and the samples of the latent classes $(\mathbf{Z}_t^{(i)})_{t=t_0, \dots, T}$ that are produced by the SEM algorithm, we can estimate

the uncertainty of the estimates. We are especially interested in the uncertainty of the confusion matrix $\hat{\Theta}$. For that we rewrite the matrix $\hat{\Theta}$ as a vector $\hat{\boldsymbol{\vartheta}}$ as defined below. Note that the matrix $\hat{\Theta}$ is of full rank, because the rows of $\hat{\Theta}$ are probability vectors that sum up to one. To account for this constraint, we drop the last column of $\hat{\Theta}$ and denote the resulting matrix as $\hat{\Theta}_{-K}$ and its rows as $\hat{\boldsymbol{\theta}}_{z,-K}$ for $z = 1, \dots, K$. Using this notation, we can define the vector $\hat{\boldsymbol{\vartheta}}$ as

$$\hat{\boldsymbol{\vartheta}} = (\hat{\boldsymbol{\theta}}_{1,-K}, \dots, \hat{\boldsymbol{\theta}}_{K,-K})^\top = (\hat{\theta}_{1,1}, \dots, \hat{\theta}_{1,K-1}, \dots, \hat{\theta}_{K,1}, \dots, \hat{\theta}_{K,K-1})^\top. \quad (11)$$

The estimate of the covariance matrix of $\hat{\boldsymbol{\vartheta}}$, denoted by $\widehat{\mathbf{V}}(\hat{\boldsymbol{\vartheta}})$, is given below. The formulas are derived in ?, chapter 10.2. and Hechinger et al. (2024).

$$\widehat{\mathbf{V}}(\hat{\boldsymbol{\theta}}_{z,-K}|Z) = \frac{\text{diag}(\hat{\boldsymbol{\theta}}_{z,-K}) - \hat{\boldsymbol{\theta}}_{z,-K} \cdot \hat{\boldsymbol{\theta}}_{z,-K}^\top}{\sum_{i=1}^n I(Z^{(i)} = z)} \quad (12)$$

$$V_t := \text{blockdiag}(\widehat{\mathbf{V}}(\hat{\boldsymbol{\theta}}_{1,-K}|Z_t), \dots, \widehat{\mathbf{V}}(\hat{\boldsymbol{\theta}}_{K,-K}|Z_t)) \quad (13)$$

$$\widehat{\mathbf{V}}(\hat{\boldsymbol{\vartheta}}) = \frac{1}{T - t_0} \sum_{t=t_0+1}^T V_t + \frac{1}{T - t_0 - 1} \sum_{t=t_0+1}^T (\hat{\boldsymbol{\vartheta}}_t - \hat{\boldsymbol{\vartheta}})(\hat{\boldsymbol{\vartheta}}_t - \hat{\boldsymbol{\vartheta}})^\top \quad (14)$$

Note that given the true latent classes Z , the probability vectors $\hat{\boldsymbol{\theta}}_{z,-K}$ and $\hat{\boldsymbol{\theta}}_{z',-K}$ are independent for $z \neq z'$. Hence, the matrix V_t in equation 14 is block-diagonal.

A Appendix

A.1 Data

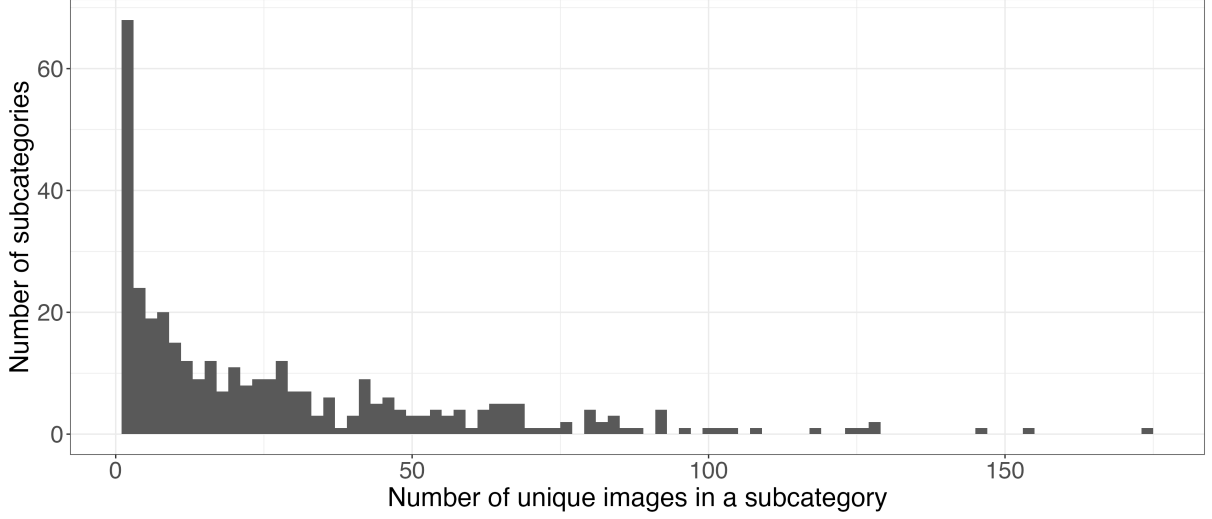


Figure 11: Distribution of the number of distinct images per subcategory. One can see that the distribution is right-skewed and that a lot of subcategories consist of only a few distinct images.

A.2 Uncertainty Estimation

We can express the MLE of the confusion matrix Θ from equation 7 using matrix notation. We define the matrix $\mathbf{Y}$ as the $n \times K$ matrix with the vote distributions $\mathbf{Y}^{(i)}$ as rows. We define the matrix $\mathbf{Z}$ as the $K \times n$ matrix with the $\mathbf{Z}^{(i)}$ as columns. Furthermore, we define the vector $\mathbf{J}$ as the $n \times 1$ vector with the $J^{(i)}, i = 1, \dots, n$ as entries. We can then write the MLE of the confusion matrix Θ as

$$\hat{\Theta} = \mathbf{Z}\mathbf{Y} \cdot \text{diag}(\mathbf{Z}\mathbf{J})^{-1} \quad (15)$$

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Munich, 27th July, 2024

Eugen Gorich